Deciphering the Hidden Code: Neural Network Insights into Cancer-Linked Transcriptional Prediction

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Abstract

Epigenetic alterations can contribute to cancer progression, as enhancement/silencing of the cancer genes through DNA methylation, histone modifications, and/or genomic imprinting may cause tumor proliferation/growth. Understanding principles behind epigenetic control of gene transcription in both standard and tumor cells is important to identify cancer-causing differences. We developed six neural network models to predict transcriptional activity (mRNA expression) from epigenetic signals (ChIP-seq interactions). We found that a model trained on the epigenetic signals of a human cell line (standard or cancerous) can predict that cell line's transcriptional activity with reasonable accuracy (79%). A model trained on multiple cell lines can be generalized to predict the transcriptional activity of a new cancerous cell line with reasonable accuracy (from 68-76%). Increasing the number of training cell lines significantly improves the model's predictive accuracy when generalizing to other cell lines. This demonstrates underlying patterns in the relationship between the epigenetic signals and resulting in
transcriptional activity, including shared patterns in normal and tumor cells. Our model is a valuable tool for exploring epigenetic landscapes from a cancer context.

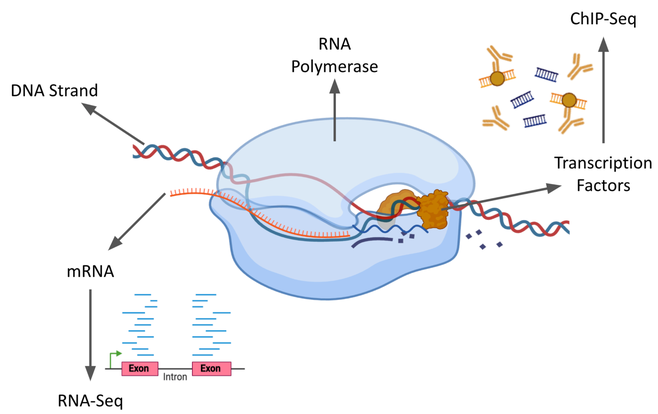


Figure 1 Chromatin Immunoprecipitation Sequencing (ChIP-seq) takes DNA cross-linked with DNA-binding proteins and creates a genome-wide map of regions where these proteins bind the DNA. RNA-seq quantifies mRNA to approximate expression rates across the genome. A method for accurately predicting RNA-seq outputs from ChIP-seq data would enable further use of existing ChIP-seq datasets and support a pattern-based understanding of the principles connecting gene regulation and transcription. *Made with BioRender*.

Results

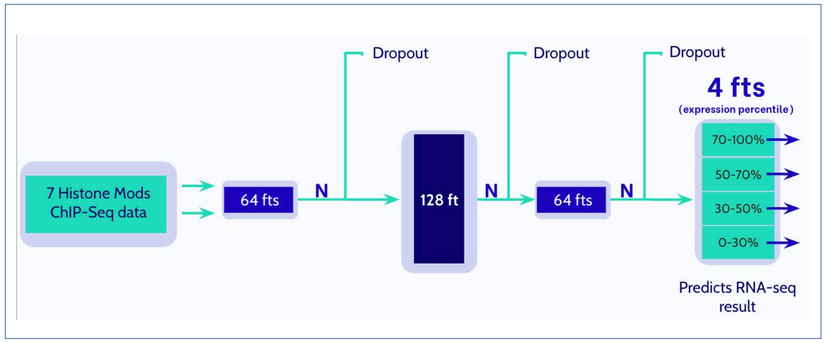
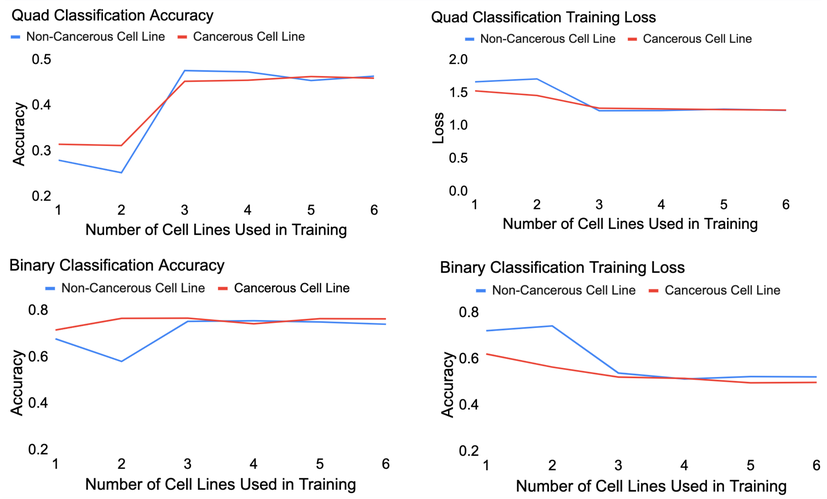


Figure 2 Schematic of the model (left). The model takes an N* × 7 input vector, with N* indicating the amount of data (an arbitrarily large number) being analyzed across seven histone modifications. For each set of layers, the model contains a dense layer with a certain number of neurons, a normalization layer, and a 10% dropout layer. The first and third set of layers have 64 neurons, while the second set has 128 neurons. The final layer creates the output (four options in the quadrant classification, two options in the binary classification).

Figure 3 Accuracy and Loss curves for the binary and quadrant classification models as the number of cell lines increases (below). The general trend indicates that accuracy increases, and loss reduces as the number of cell lines increases. A Mann-Whitney U test was used because the data fails the normality assumption. In all four curves the p-value is substantially greater than 0.05, so we are 95% confident that differences in the curves are due to random chance. This is expected because it indicates that the model works similarly on different kinds of cell lines, so it is properly generalizable across cancerous and non-cancerous cell lines.



Receiver Operating Characteristic (ROC) Curves with N cell lines

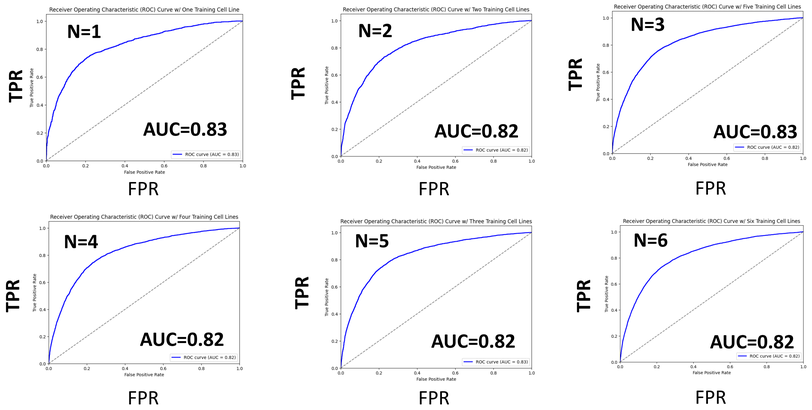


Figure 4 ROC curves with the corresponding AUC values for each number of cell lines (above). The AUC stays relatively constant regardless of the number of cell lines, indicating that the model architecture works well regardless of the number of cell lines.

Methods

We accessed ChIP-seq and RNA-seq data from the ENCODE Project. We used 64 call sets[1] (Bernstein et al. 2011; Bernstein et al. 2012; Bernstein et al. 2016; Bernstein et al. 2017; Ren et al. 2013; Wold et al. 2021; Wold et al. 2022; Gingeras et al. 2014; Reddy et al. 2022) (www.encodeproject.org/). Using this data from six unique cell lines, we trained and validated a model on four of the cell lines (one non-cancerous cell line and three cancerous cell lines) and tested the model on two of the cell lines (one non-cancerous cell line and one cancerous cell line).

Conclusions

Overall, this study demonstrates that our model can accurately predict transcriptomic activity from epigenetic signals. Our finding that increasing the number of cell lines generally increases the accuracy of the model indicates that there is a distinct link between epigenetic signal and transcriptomic activity that exists regardless of cell line. This model could be used to accurately predict transcriptional activity of any cell line if only epigenetic data is available. Moreover, the patterns and weights the model assigns could be used to further describe the regulatory rules that control gene transcription and, by extension, cellular function. To continue improving this model, we will add data from more cell lines and try to label the data with its corresponding cell line so that the model can more accurately assign weights, improving the overall accuracy. We will also try experimenting with deeper model architectures.

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References

1. ENCFF190RZM, ENCFF701JXT, ENCFF480KNX, ENCFF087YCU, ENCFF691LFQ, ENCFF174RRQ, ENCFF688HLG, ENCFF148PNW, ENCFF400CAD, ENCFF811ZFE, ENCFF907GKJ, ENCFF045HZZ, ENCFF733CJA, ENCFF682HNO, ENCFF457URZ, ENCFF054RSU, ENCFF767UON, ENCFF465GBD, ENCFF582IMB, ENCFF559MMQ, ENCFF239EBH, ENCFF182EJH, ENCFF624KAH, ENCFF213WKK, ENCFF277XII, ENCFF232QSG, ENCFF572IBD, ENCFF558LSB, ENCFF554XSR, ENCFF767KKA, ENCFF219ZOU, ENCFF259DOA, ENCFF257BOE, ENCFF125NHB, ENCFF192NNA, ENCFF905WYC, ENCFF230LCP, ENCFF764XQX, ENCFF730VLE, ENCFF786RCE, ENCFF173DKI, ENCFF517WOS, ENCFF763NCP, ENCFF442RRY, ENCFF163MXP, ENCFF138YNG, ENCFF163QKN, ENCFF481DZL, ENCFF327XJC, ENCFF843TBP, ENCFF366QUC, ENCFF496OWV, ENCFF175EDR, ENCFF612JOH, ENCFF807MNG, ENCFF499QYJ, ENCFF409USY, ENCFF718UDS, ENCFF730LJD, ENCFF584RWQ, ENCFF742JXC, ENCFF035MLG, ENCFF992CSD, ENCFF397MAI